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# A new pathogen pattern of acute respiratory tract infections in primary care after COVID-19 pandemic: a multi-center study in southern China

Heng Zhang[1†], Deqi Meng[2†], Hao Huang[3†], Longhao Feng[2], Yushao Li[4], Yong Jiang[5], Ling Wang[6], Ruoyu Deng[7], Yuchun Sun[8], Binqin Chen[9], Fuzhao Liao[10], Yumei Wu[11], Huancai Zheng[12], Junling Ding[13], Maokun Chen[14], Cui Zeng[15], Wanting Zhao[16], Meng Hou[17], Yinyan Li[18], Zhishen Li[19], Haibo Xia[20], Kai Yang[1*] and Lingwei Wang[1*]

## Abstract

**Background** After the coronavirus disease 2019 (COVID-19) pandemic, no studies on bacterial and atypical pathogens were conducted in primary care. We aimed to describe the etiological composition of acute respiratory tract infections (ARTIs) presenting to primary care with limited resources after the pandemic.

**Methods** 1958 adult patients with ARTIs from 17 primary care clinics were recruited prospectively from January 2024 to March 2024. 17 and 62 pathogens in throat swab samples were tested using polymerase chain reaction (PCR) and targeted next-generation sequencing (tNGS), respectively. We analyzed the pathogen spectrum and co-infectious pattern of viral, bacterial or atypical pathogens. Then, the associations between clinical characteristics and pathogens were investigated.

**Results** In PCR test, the positive rate of any pathogens was 80.3%, consisting of 60.2% for viruses, 41.8% for bacteria and 21.7% for viral-bacterial co-infection. In tNGS test, the positive rate was 89.1%, consisting of 64.7% for viruses, 55.2% for bacteria and 30.9% for viral-bacterial co-infection. Influenza virus B (18.2%), influenza virus A (16.8%) and severe acute respiratory syndrome coronavirus 2 (14.1%) were the three leading viral pathogens, and *H. influenzae* (36.1%), *S. anginosus* (15.7%) and *S. pneumoniae* (8.4%) were the three leading bacterial pathogens. Few *M. pneumoniae* (1.6%) were detected. The mixed bacterial or mixed viral-bacterial co-infections were the most common co-infectious patterns. The mixed bacterial or mixed viral-bacterial co-infections were the most common co-infectious patterns. Overall, patients with viral infection or viral-bacterial co-infection had more clinical symptoms, and patients with bacterial infection had higher inflammatory indicators.

[†]Heng Zhang, Deqi Meng and Hao Huang contributed to this work equally.

*Correspondence:
Kai Yang
yang.kai@szhospital.com
Lingwei Wang
wang.lingwei@szhospital.com

Full list of author information is available at the end of the article

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**Conclusions** After the COVID-19 pandemic, the main viral pathogens of ARTIs were unevenly distributed, and less bacterial and atypical pathogens were detected in primary care. The microbiological evidences can optimize the precision diagnosis and treatment of ARTIs in primary care with limited resources.

**Keywords** Respiratory tract infections, Primary health care, Pandemics, Virus infections, Bacterial infections, Coinfection

## Introduction

Acute respiratory tract infections (ARTIs) are the most common infectious disease worldwide, especially in the underdeveloped and developing countries [1]. The etiological agents for ARTIs include bacteria, viruses and co-infection, in which viral pathogens are the major cause of ARTIs, such as rhinovirus, influenza virus, respiratory syncytial virus and parainfluenza virus [2, 3]. During the coronavirus disease (COVID-19) pandemic, numerous preventive strategies for COVID-19 have also significantly decreased other respiratory infections [4, 5]. After the gradual easing of restrictions, the co-occurrences of COVID-19 resurgence and other ARTIs have been a new challenge to the primary care system [6]. Therefore, re-evaluating etiological composition can describe the new pathogen spectrum and promote precision therapy of ARTIs in primary care settings.

Primary care clinics are the first selection for most patients with ARTIs. Due to the limited resources, including the challenges in clinical sampling, limited diagnostic tools and lack of efficacy evaluation after empirical treatment, it is usually difficult for general practitioners to figure out the etiology of ARTIs, leading to considerable unnecessary antibiotic use and a waste of health-care costs [7, 8]. To date, few studies on the etiological and epidemiological features of respiratory tract infections have been conducted in primary care clinics, which were all focused on acute lower respiratory tract infections (ALRTIs). In a prospective study from 11 European countries, bacterial and viral pathogens were identified in one in five and less than half of adult patients with ALRTIs in primary care, which provided microbiological evidence for the failure of empiric antibiotic therapy [9]. In another study in the UK, 26% and 63% of 80 adult patients with ALRTIs presenting to primary care were caused by bacterial and viral pathogens [10]. Numerous previous studies have also reported the etiology of patients with ARTIs from secondary and tertiary teaching hospitals [11–13]. However, these studies lacked generalizability for primary care settings, especially after the end of the COVID-19 pandemic.

The primary objective of this study was to describe the etiological composition of patients with ARTIs from 17 primary care clinics in Shenzhen, a city in southern China. Our secondary objective was to compare the results of pathogen detection by polymerase chain reaction (PCR) and targeted next-generation sequencing (tNGS).

## Methods

### Patients and clinical specimens

This prospective study recruited participants between January 12, 2024 and March 19, 2024 by 153 primary care practitioners from 17 primary care clinics in Shenzhen, China (Figure S1). The inclusion criteria of this study were as follows: (1) 14 years old and above; (2) disease onset within 7 days; (3) fever (≥37.3 ℃) combined with respiratory symptoms (cough, sore throat, nasal congestion or runny nose); (4) agree to collect throat swabs and provide written informed consent before enrolment. Information on age, sex, clinical symptoms, blood routine and C-reactive protein (CRP) were collected using a self-administered questionnaire (Supplementary Methods). Two throat swabs were collected by trained technicians, transported to the laboratory at 4 °C and subjected to both PCR and tNGS within 24 h of collection simultaneously.

### PCR for 17 pathogens

Throat swab samples were sent to a commercial company laboratory for the detection of 17 pathogens by multiplex PCR with capillary electrophoresis. These included 10 viruses, 5 bacteria, 1 mycoplasma, and 1 chlamydia (Supplementary Table 1).

#### *Sample preparation*

Add an appropriate amount of saline or PBS to the sampling tube and shake well.

#### *Nucleic acid extraction*

The sample was extracted with 400 μ L of the test sample through the Nucleic Acid Extraction or Purification Kit (Ningbo Haier Gene Technology Co., Ltd.). In each well, 10 μ L of proteinase K and 2 μ L of RT-PCR internal reference were added in advance.

#### *Amplification*

The PCR amplification was performed with 5 μ L of nucleic acid using a pathogen multiplex kit (Ningbo Hirsch Gene Technology Co., Ltd.).

#### *Separation of PCR products by capillary electrophoresis*

Electrophoresis was performed with 9 μ L of electrophoresis sample buffer (2.5%, SIZE-500 Plus in Hi Di ™) + 1 μ L of PCR product. Sequencing was performed with 3500xl-Dx.

#### *Bioinformatics analysis*

GeneMapper was used to analyze the results and identify pathogenic species.

### Targeted next-generation sequencing for 62 pathogens

Another throat swab samples were sent to the same laboratory for tNGS analysis for 62 pathogens simultaneously, including 30 viruses, 24 bacteria and 5 mycoplasmas and 3 chlamydia (Supplementary Table 2).

#### *Sample preparation*

The sample was mixed with 1300 μ L of 0.1 M DTT and allowed to stand for 3–5 min until the sample was liquefied (not necessary for sputum or viscous alveolar lavage, clear alveolar lavage, or throat swabs for nucleic acid preservation). The supernatant was centrifuged and discarded (12000 r/min, 5 min). The precipitate was taken for wall-breaking operation. Subsequently, the supernatant was centrifuged and discarded (12000 r/min, 5 min).

#### *Nucleic acid extraction*

500 μL of the homogenate was utilized for total nucleic acid extraction and purification via the MagPure Pathogen DNA/RNA Kit (R6672-01B, Magen, Guangzhou, China).

#### *Library construction*

The cDNA was reverse transcribed from 14 μ L of nucleic acid using the Respiratory Pathogen Targeted Sequencing Kit (KingCreate, Guangzhou, China). The PCR product was centrifuged and purified by adding 25 μ L of magnetic beads. 11.5 μ L of the purified product was used for library construction. The PCR product was purified by repeating the above steps. Qubit was used to measure the library concentration for quality control. The sample libraries were pooled and the total nucleic acid concentration and fragment size were measured.

#### *Sequencing*

The diluted and denatured library was subsequently subjected to sequencing on an Illumina MiniSeq platform using a universal sequencing reagent kit (KS107-CXR, KingCreate, Guangzhou, China).

#### *Bioinformatics analysis*

Sequencing data were analyzed using the data management and analysis system (v3.7.2, KingCreate). The raw data underwent initial identification via the adapter. Reads with single-end lengths exceeding 50 bp were retained, followed by low-quality filtering to retain reads with Q30 > 75%, ensuring high-quality data. The single-ended aligned reads were then compared using the Self-Building clinical pathogen database to determine the read count of specific amplification targets in each sample. The reference sequences used for read mapping were a database curated from different sources, including the Genbank database, Refseq database, and Nucleotide database from NCBI.

### Statistical analysis

Baseline characteristics were displayed as mean (standard deviation, SD) for continuous variables and number (percentage) for categorical variables. The difference between rates was evaluated by chi-square test. The results of PCR and tNGS were compared by the McNemar test and kappa coefficient. The kappa values were classified into poor (< 0.00), mild (0.00–0.20), relatively good (0.21–0.40), fair (0.41–0.60), good (0.61–0.80) and excellent (0.81–1.00) agreement [14]. All statistical analyses were performed using SAS 9.4 (SAS Institute Inc., Cary, North Carolina), and a two-sided $P$ value < 0.05 was considered statistically significant unless specified otherwise.

## Results

### Demographic characteristics

In total, 2028 patients with ARTIs were recruited and had samples tested, from which 70 cases were excluded owing to incomplete data and inclusion criteria. Therefore, 1958 patients were included in the final analyses of this study (Fig. 1). The baseline data were shown in Table 1. The average temperature was 38.2 ℃ and the median disease duration was 1 day. The common symptoms were sore throat (82.5%), cough (80.9%), fatigue (70.2%), headache (64.6%), nasal obstruction (64.4%), running nose (63.3%) and muscle pain (59.1%). Among patients with blood cell count information, 27.6%, 41.3% and 17.2% had higher levels of white blood cell (WBC), Neutrophil (NEU) and lymphocyte (LYM), respectively. 64.6% of patients also had increased CRP levels.

### Pathogen spectrum

In the PCR test, 80.3% of patients with ARTIs tested positive for at least one pathogen, in which the positive rate of viruses was 60.2%, the positive rate of bacteria was 41.8%, and the positive rate of viral-bacterial co-infection was 21.7% (Fig. 2A). Based on the proportion of positive detection for viruses, the most commonly determined viral pathogens were influenza virus B (IFV-B, 18.2%), influenza virus A (IFV-A, 15.9%) and severe acute respiratory syndrome coronavirus 2 (SARS-COV-2, 14.1%), followed by human rhinovirus (HRV, 8.3%), human adenovirus (HAdV, 2.6%) and human metapneumovirus

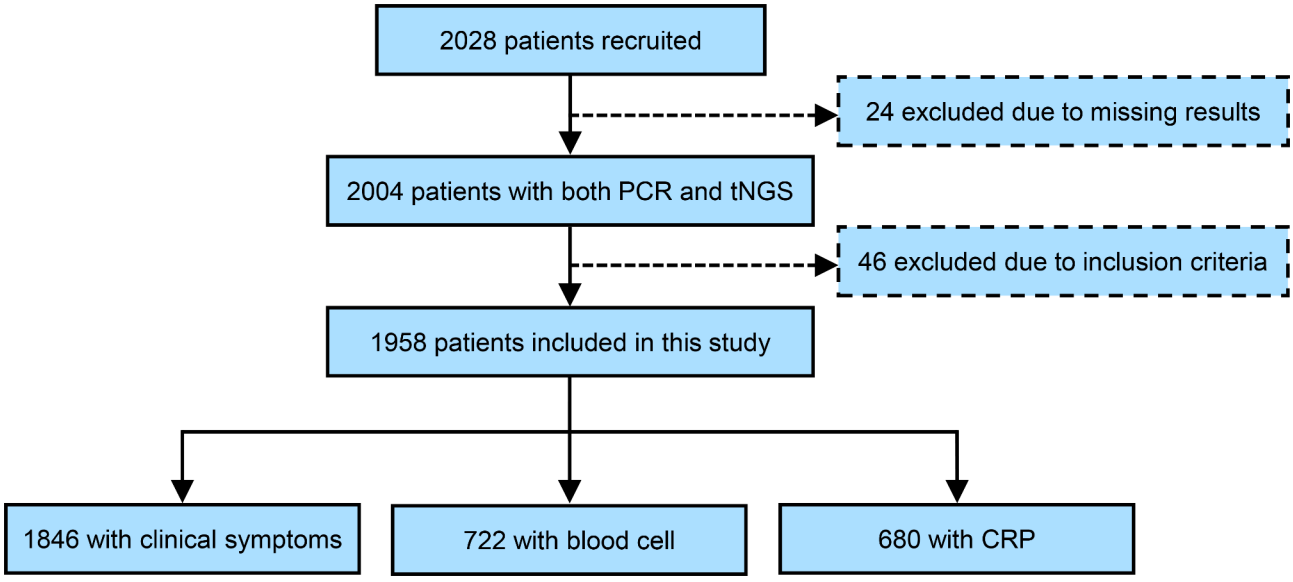


**Fig. 1** The flowchart and sample size of patient in different analyses in this study

**Table 1** Baseline characteristics of included patients

| Demographics | Statistics |
|---|---|
| Age, year | 35.1 ± 11.7 |
| Male, % | 904(46.2) |
| Disease related characteristics | |
| Temperature, °C | 38.2 ± 0.6 |
| Disease duration, day | 1.0(1.0, 2.0) |
| Running nose, % | 1169(63.3) |
| Sore throat, % | 1523(82.5) |
| Nasal obstruction, % | 1189(64.4) |
| Cough, % | 1493(80.9) |
| Fatigue, % | 1296(70.2) |
| Headache, % | 1193(64.6) |
| Muscle pain, % | 1091(59.1) |
| Loss of taste, % | 153(8.3) |
| Olfactory loss, % | 111(6.0) |
| Nausea, % | 232(12.6) |
| Vomiting, % | 85(4.6) |
| Diarrhea, % | 84(4.6) |
| WBC ≥ $10 \times 10^9$/L | 199(27.6) |
| NEU ≥ $7 \times 10^9$/L | 291(41.3) |
| LYM ≥ $4 \times 10^9$/L | 118(17.2) |
| CRP ≥ 5 mg/L | 439(64.6) |

(HMPV, 1.3%) (Table 2). *H. influenzae* and *S. pneumoniae* were the most frequent bacterium pathogens, accounting for 36.1% and 8.4% of the total positive detection.

In the tNGS test, the positive rate of any pathogens was 89.1%, consisting of 64.7% for viruses, 55.2% for bacteria and 30.9% for viral-bacterial co-infection (Fig. 2B). The results of most pathogens showed good/excellent concordances between the two methods (kappa coefficient > 0.6), while the concordances of human coronavirus (HCoV), *B. pertussis* and *S. pneumoniae* were lower (Table 2). In addition, tNGS has detected more pathogens, in which the top-listing pathogens were *S. anginosus* (15.7%), *S. aureus* (5.6%), *K. pneumoniae* (5.5%), *M. katamorella* (3.9%) and *S. pyogenes* (2.3%), Epstein-Barr virus (EBV, 1.8%) and herpes simplex virus type 1 (HSV-1, 1.6%) (Supplementary Table 3).

### Co-infectious pattern

In PCR test, co-infection of two or more pathogens was observed in 25.1% of patients with ARTIs (Fig. 3A). The mixed bacterial or mixed viral-bacterial co-infections were the most frequent co-infectious patterns (Fig. 3C). *H. influenzae* was frequently co-infected with other pathogens, followed by *S. pneumoniae*. Typical respiratory pathogens were more frequently co-infected with viruses than atypical pathogens (Supplementary Table 4).

In tNGS test, 41.5% of patients with ARTIs were infected by at least two pathogens due to the detection of more pathogens (Fig. 3B). Similar to those in PCR tests, dual infection was most frequently caused by *H. influenzae* and other pathogens (Fig. 3D). *S. anginosus* was also frequently co-infected with IFV-A and IFV-B. The co-infection of *S. pneumoniae* with other pathogens was less observed in tNGS test because of the significantly lower positive rate.

The male patients with viral infection were more prone to complications with bacterial infection (Supplementary Table 5). Higher temperature, runny nose, sore throat, nasal congestion, cough, fatigue, headache, muscle pain and loss of taste were clinical symptoms of viral infection or viral-bacterial co-infection. As disease duration increased, the positive rate of viral pathogens was decreased and the positive rate of bacterial pathogens was increased, leading to more viral-bacterial

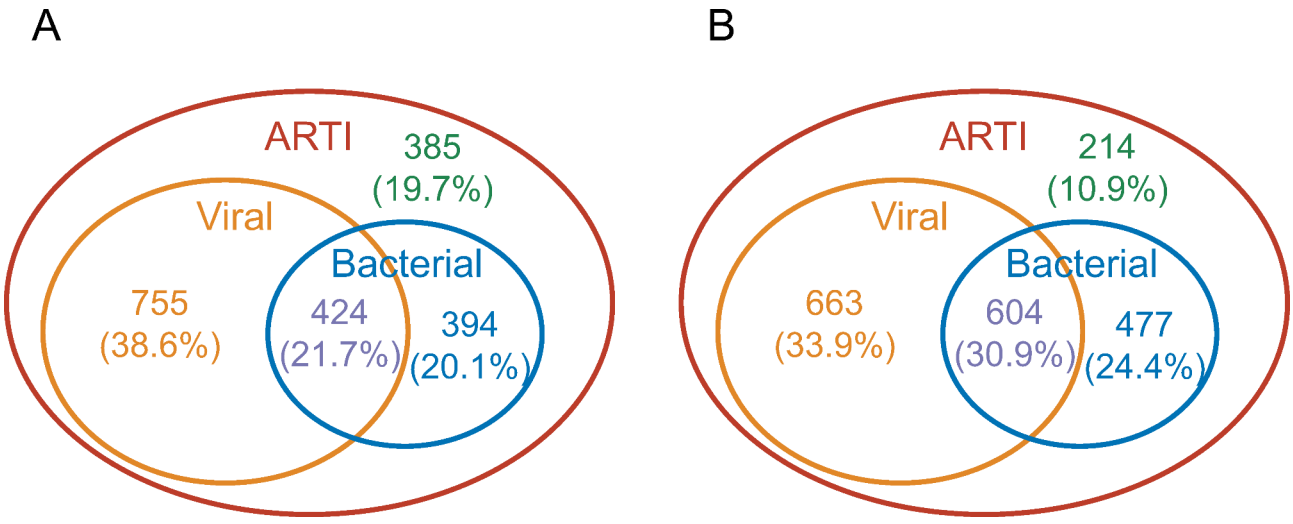


**Fig. 2** Venn diagrams of numbers (percentages) of patients with no, a viral, a bacterial or a mixed viral-bacterial etiology. (**A**) Detected by PCR. (**B**) Detected by tNGS

**Table 2** The positive rates of respiratory pathogens detected by both PCR and tNGS

| Pathogens | PCR | tNGS | Kappa | *P* |
|---|---|---|---|---|
| Viruses | | | | |
| Human adenovirus (HAdV) | 50(2.6) | 62(3.2) | 0.78 | 0.01 |
| Human bocavirus (HBoV) | 0(0.0) | 0(0.0) | - | - |
| Human coronavirus (HCoV) | 12(0.6) | 26(1.3) | 0.58 | < 0.01 |
| Human metapneumovirus (HMPV) | 25(1.3) | 29(1.5) | 0.81 | 0.21 |
| Human parainfluenza virus (HPIV) | 13(0.7) | 21(1.1) | 0.76 | < 0.01 |
| Human rhinovirus (HRV) | 162(8.3) | 198(10.1) | 0.82 | < 0.01 |
| Influenza virus A (IFV-A) | 329(16.8) | 312(15.9) | 0.95 | < 0.01 |
| Influenza virus B (IFV-B) | 355(18.1) | 357(18.2) | 0.95 | 0.72 |
| Respiratory syncytial virus (RSV) | 10(0.5) | 20(1.0) | 0.66 | < 0.01 |
| SARS-COV-2 | 276(14.1) | 275(14.0) | 0.98 | 0.74 |
| Any of the above viruses | 1179(60.2) | 1226(62.6) | 0.86 | < 0.01 |
| Bacteria | | | | |
| *B. pertussis* | 6(0.3) | 15(0.8) | 0.57 | < 0.01 |
| *C. pneumoniae* | 1(0.1) | 2(0.1) | 0.67 | 0.32 |
| *H. influenzae* | 707(36.1) | 689(35.2) | 0.68 | 0.29 |
| *L. pneumophila* | 0(0.0) | 0(0.0) | - | - |
| *M. pneumoniae* | 18(0.9) | 32(1.6) | 0.68 | < 0.01 |
| *M. tuberculosis* | 0(0.0) | 0(0.0) | - | - |
| *S. pneumoniae* | 164(8.4) | 74(3.8) | 0.28 | < 0.01 |
| Any of the above bacteria | 818(41.8) | 757(38.7) | 0.61 | < 0.01 |
| Any of the above pathogens | 1573(80.3) | 1604(81.9) | 0.67 | 0.03 |

co-infection in patients with intermediate disease duration. Higher WBC, NEU and CRP were inflammatory indicators for bacterial infection.

### Main pathogen detection according to clinical characteristics

In elderly patients, the positive rates of HRV, SARS-COV-2, *S. pneumoniae*, *S. anginosus* were higher, and the positive rate of IFV-B was lower than those in adult patients (Supplementary Table 6). Male patients had higher positive rates of *H. influenzae and K. pneumoniae*. The infection of IFV-A and IFV-B were associated with higher body temperature, and HRV and *S. anginosus* infection was associated with lower body temperature. The patients with IFV-A, IFV-B and SARS-COV-2 had shorter disease duration, while those with *H. influenzae* and *S. anginosus* had longer disease duration.

Patients with IFV-A, IFV-B or SARS-COV-2 infection had similar clinical manifestations, including higher incidences of runny nose, nasal congestion, cough, fatigue,

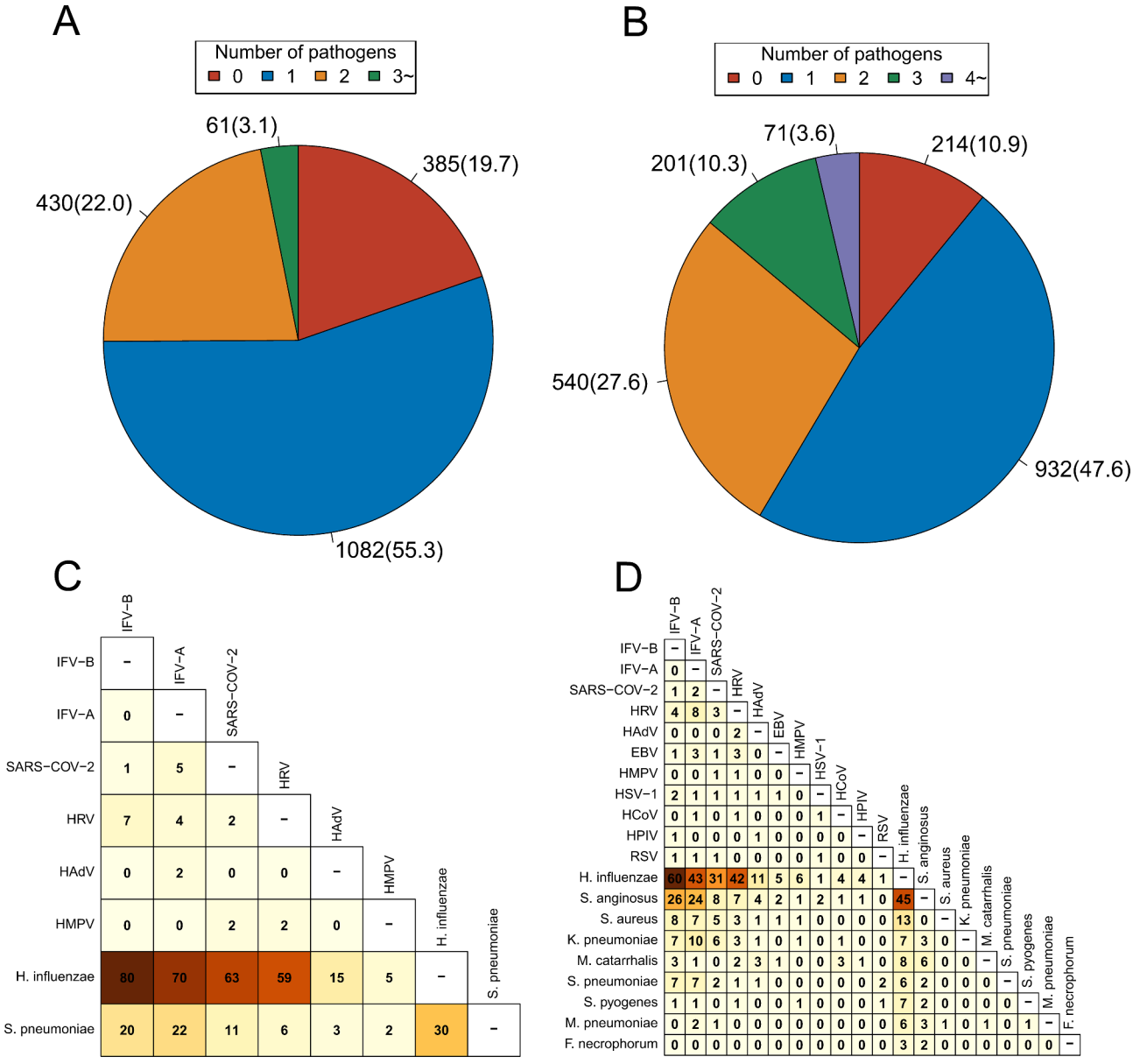


**Fig. 3** The detailed co-infectious patterns of main pathogens. (**A**) The number of pathogens detected by PCR. (**B**) The number of pathogens detected by tNGS. (**C**) The number of patients infected by two pathogens in PCR. (**D**) The number of patients infected by two pathogens in tNGS

headache, muscle pain, loss of taste and olfactory loss. HRV infection was associated with higher incidences of runny nose, nasal congestion and cough, and lower incidences of fatigue, headache, muscle pain, nausea and vomiting. Patients with positive bacteria had less clinical manifestation compared to those with positive viruses. Patients with elevated WBC had higher positive rates of HRV and *H. influenzae*, and lower positive rates of IFV-A, IFV-B and SARS-COV-2, which were mainly caused by elevated NEU. Elevated CRP was also associated with a higher incidence of bacterial infection, and a lower positive rate of IFV-B.

## Discussion

Using PCR and tNGS, we exhibited the viral and bacterial pathogen spectrum of ARTIs in primary care in the first flu season after the COVID-19 pandemic in southern China. We found that the main viral pathogens were more unequally distributed compared with those in previous studies, and fewer bacterial and atypical pathogens were detected. The positive rate and co-infectious pattern were affected by clinical characteristics. We also found similar results of PCR and tNGS.

Few previous studies on respiratory pathogens of ARTIs have focused on mildly and moderately ill patients in primary care. Between 2007 and 2010, a study reported

nine viral pathogens and six bacterial pathogens of ALRTIs in adults in 16 primary care networks from 11 European countries [9]. Between 2015 and 2017, a surveillance study in France on patients with influenza-like illness in primary care reported positive IFV, respiratory syncytial virus (RSV), HMPV and HRV [15]. Compared with these studies, our study detected more pathogens, used both PCR and tNGS, and was the first to be affected by COVID-19, which accounted for 14% of viral infections.

The viral pathogen spectrum in our study was different from other studies before the easing of COVID-19 pandemic restrictions in China. Compared with the nationwide surveillance of pathogens on patients with ARTIs by the Chinese Center for Disease Control and Prevention between 2009 and 2019, our study had a higher positive rate of IFV-B, and lower positive rates of most other viral pathogens, including IFV-A, HRV, HAdV, RSV, human parainfluenza virus (HPIV), HMPV, and so on [11]. A similar result was also observed when compared with other studies before and during the COVID-19 pandemic restriction [2, 16, 17].

A few explanations can account for the significant change in the viral pathogen spectrum. The infection of SAS-COV-2 may contribute to the low positivity of viral pathogens, as previous studies reported that the co-infection of SARS-COV-2 and other respiratory viruses was uncommon [17, 18]. The non-pharmaceutical preventive interventions for SARS-COV-2 have reduced the prevalence of other respiratory viruses and virus diversity, leading to variability in the return of some respiratory viruses after the pandemic [19]. The patients presenting to primary care often consulted doctors within the first few days of their illness and had milder clinical symptoms, which were different from those in high-level hospitals [20]. The population in this study was more representative of the primary care population than previous studies by the Center for Disease Control and high-level hospitals, who were more likely to be of lower socio-economic status, elderly, and individuals with chronic conditions. The study design, inclusion criteria and diagnostic methods may also explain the different viral pathogen spectrum.

As for bacterial pathogens, the positive rate of *H. influenzae* was much higher, and the positive rate of other bacterial pathogens was lower than those in most previous studies [11, 21]. The low bacterial positive rate was also reported in the study based on primary care clinics in Europe [9]. Generally, the patients with ARTIs firstly sought medical care from general practitioners, resulting in a generally shorter disease course for patients in primary care clinics [22, 23]. We also found that patients with shorter disease course had a lower proportion of bacterial infection. Furthermore, viral ARTIs with longer disease courses had a higher risk of secondary bacterial infections [24, 25]. Therefore, the shorter disease course and lower risk of secondary bacterial infections may explain the lower positive rate of bacterial pathogens in our study and other studies in primary care. These results provided microbiological evidence that antibiotics may not be beneficial for some patients with ARTIs and primary care practitioners should pay more attention to rationalizing antibiotic prescribing for ARTIs.

The positive rate of *M. pneumoniae* in this study was 1.6%, which was much lower than those of other studies during the same period [26–28]. Recent studies reported that the current outbreak of *M. pneumoniae* infection was predominantly caused by macrolide-resistant *M. pneumoniae*, which often manifested as severe symptoms, longer disease course and poorer prognosis [29, 30]. These patients tended to visit high-level hospitals for outpatient and inpatient services. Another study from a tertiary hospital in the same city as ours reported a positive rate of around 30% for *M. pneumoniae* [31]. Additionally, *M. pneumoniae* has a long incubation period of more than 10 days. We also found 15 (0.8%) adult patients with positive *B. pertussis* in this study, which was easily overlooked and misdiagnosed. This result indicated that the burden of *B. pertussis* in adults may be underestimated in the current policy of passive surveillance [32].

We firstly compared PCR and tNGS in detecting the etiology of ARTIs. Compared with PCR, tNGS can detect more potential pathogens simultaneously [33]. The quantitative result of tNGS could also provide more insights into pathogen information [33]. However, the urgent task of primary care clinics is to identify several common pathogens with a simple, rapid and cheap method. In such cases, PCR is more feasible and cost-efficient.

The tNGS method detected less *S. pneumoniae* than PCR. We speculate that there are two possible scenarios. Firstly, there are differences between the two methods. PCR amplifies only one segment of the nucleic acid sequence, whereas tNGS amplifies more segments of the nucleic acid sequence. In addition, it is possible that the primers used in the PCR kit are not as specific as the tNGS kit. Therefore, when pathogens of the genus *Streptococcus* are present in the sample, tNGS amplifies and recognizes a wide range of *streptococci* such as *S. pneumoniae*, *S. pyogenes*, and *S. anginosus*. Whereas PCR methodology may identify pathogens of the genus *Streptococcus* as *S. pneumoniae* after amplification. Secondly, the two samples contain different nucleic acids. Although the time interval was short, the nucleic acids of the pathogens taken from the two throat swabs may be different. In addition, there are differences in the nucleic acid extraction reagents (nucleic acid extraction process) used in the two methods, which may lead to differences in the purified nucleic acids.

The current study has several limitations. First, the data for clinical symptoms, blood cells and CRP were not collected for all patients, which may lower the statistical power of studying their relationships to pathogens. Second, we did not collect information about antibiotic usage and chest X-ray, which may underestimate the prevalence of bacterial pathogens in the respiratory tract and could not distinguish between upper and lower respiratory tract infections. Third, this study was performed in 17 primary care clinics in a city in southern China, therefore limiting the generalizability of the findings. Last, this study only focused on the viral and bacterial pathogen spectrum in the first flu season after the COVID-19 pandemic, which cannot evaluate the seasonal variations.

## Conclusions

In conclusion, the current study provided a quite different viral pathogen spectrum among adult patients in primary care after the COVID-19 pandemic, mainly composed of IFV-B, IFV-A and HRV. Less bacterial and atypical pathogens were detected, except for *H. influenzae*. These microbiological evidence would inform primary care practitioners for precision diagnosis and therapy of ARTIs.

### Abbreviations

| | |
|---|---|
| ALRTIs | Acute lower respiratory tract infections |
| ARTIs | Acute respiratory tract infections |
| COVID-19 | Coronavirus disease 2019 |
| CRP | C-reactive protein |
| EBV | Epstein-Barr virus |
| HAdV | Human adenovirus |
| HCoV | Human coronavirus |
| HMPV | Human metapneumovirus |
| HPIV | Human parainfluenza virus |
| HRV | Human rhinovirus |
| HSV-1 | Herpes simplex virus type 1 |
| IFV-A | Influenza virus A |
| IFV-B | Influenza virus B |
| LYM | Lymphocyte |
| NEU | Neutrophil |
| PCR | Polymerase chain reaction |
| RSV | Respiratory syncytial virus |
| SARS-CoV-2 | Severe acute respiratory syndrome coronavirus 2 |
| SD | Standard deviation |
| tNGS | Targeted next-generation sequencing |
| WBC | White blood cell |

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12879-025-10500-w.

Additional file 1: **Figure S1**: The geographical distribution of 17 primary care clinics in this study. **Table S1**: 17 pathogens in PCR. **Table S2**: 62 pathogens in tNGS. **Table S3**: The positive rates of other pathogens detected by tNGS. **Table S4**: The co-infection of viruses with typical and atypical respiratory pathogens. **Table S5**: The relationship between clinical characteristics and co-infection pattern. **Table S6**: The positive of main pathogens according to clinical characteristics

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Not applicable.

### Author contributions

L.W. and K.Y. designed the study. H.Z. and K.Y. analyzed and interpreted the data. H.Z. and L.W. drafted the manuscript. D.M., H.H., L.F., Y.L., Y.J., L.W., R.D., Y.S., B.C., F.L., Y.W., H.Z., J.D., M.C., C.Z., W.Z., M.H., Y.L., Z.L. and H.X. included patients and collected data. All authors revised and approved the manuscript.

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### Data availability

The datasets generated and/or analysed during the current study are available in the Zenodo repository (https://doi.org/10.5281/zenodo.14004650).

## Declarations

### Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethical Review Committee at Shenzhen People's Hospital (LL-KY-2024127-02). At recruitment, all participants gave informed consent to participate.

### Consent for publication

Not applicable.

### Competing interests

The authors declare no competing interests.

### Clinical trial number

Not applicable.

### Author details

[1]Department of Pulmonary and Critical Care Medicine, Shenzhen Institute of Respiratory Diseases, Shenzhen People's Hospital, the Second Clinical Medical College, Jinan University; the First Affiliated Hospital, Southern University of Science and Technology, Shenzhen 518001, China
[2]Nanling Community Health Service Center, the Fourth People's Hospital of Shenzhen Longgang District, Shenzhen 518001, China
[3]Community Health Management Office, Longgang District Sixth People's Hospital, Shenzhen 518001, China
[4]Nanlian Community Health Service Center, Shenzhen Longgang Central Hospital, Shenzhen 518001, China
[5]Department of Respiratory Medicine, Shenzhen Hospital of Integrated Traditional Chinese and Western Medicine, Shenzhen 518001, China
[6]Maanshan Community Health Service Center, Shenzhen Hospital of Integrated Traditional Chinese and Western Medicine, Shenzhen 518001, China
[7]Minle Community Health Service Center, the People's Hospital of Longhua Shenzhen, Shenzhen 518001, China
[8]Buchong Community Health Service Center, Shenzhen Integrated Traditional Chinese and Western Medicine Hospital, Shenzhen 518001, China
[9]Jingxin Community Health Service Station, Shenzhen Hospital(Futian) of Guangzhou University of Chinese Medicine, Shenzhen 518001, China
[10]Julong Garden Community Health Service Center, Shenzhen Pingle Orthopedics Hospital (Pingshan District Traditional Chinese Medicine Hospital), Shenzhen 518001, China
[11]Yadi Sancun Community Health Service Center, Shenzhen Pingle Orthopedics Hospital (Pingshan District Traditional Chinese Medicine Hospital), Shenzhen 518001, China
[12]Tiandong Community Health Service Center, Yantian District People's Hospital, Shenzhen 518001, China
[13]Sanxigaoyuan Community Health Service Center, Shenzhen Dapeng New District Medical and Health Group, Shenzhen 518001, China

[14]Liantang Subdistrict Community Health Center, Shenzhen Luohu Hospital Group, Shenzhen 518001, China
[15]University Town East Community Health Service Center, Shenzhen Nanshan Medical Group Headquarters, Shenzhen 518000, China
[16]Longwei Community Health Service Center, the Second People's Hospital of Futian District Shenzhen, Shenzhen 518000, China
[17]Kuichong Community Health Service Center, Kuichong People's Hospital, Shenzhen 518000, China
[18]Xiaomeisha Community Health Service Center, Shenzhen Yantian District People's Hospital, Shenzhen 518000, China
[19]Cuizhu Community Health Service Center, Shenzhen Luohu Hospital Group, Shenzhen 518001, China
[20]Xinwei Community Health Service Center, Shenzhen Nanshan Medical Group Headquarters, Shenzhen 518000, China

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